

5. Sam carries out an experiment to determine the mass of a metal block using the principle of moments.

(a) State the principle of moments.

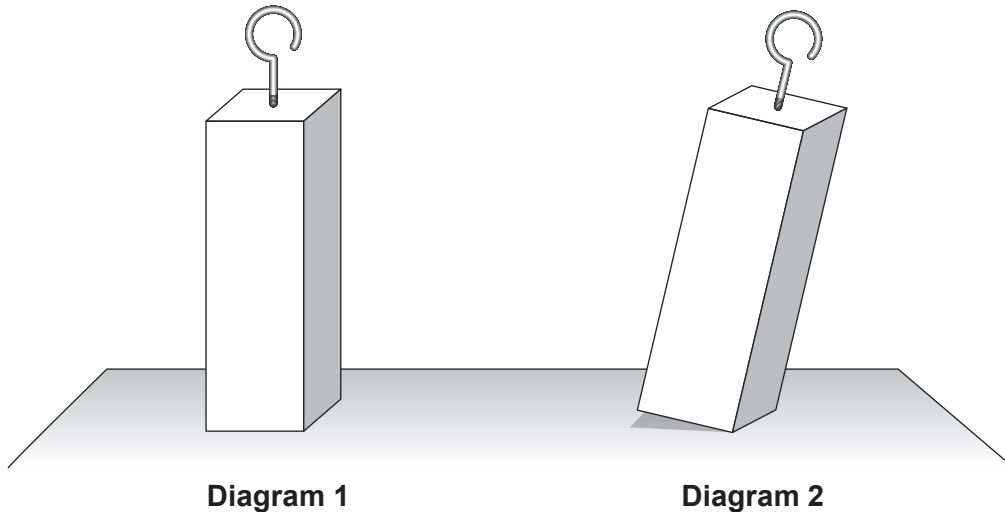
[2]

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- (b) The metal block is in the form of a cuboid as shown in **Diagram 1**. When setting up the experiment Sam accidentally knocks the metal block so that it tilts to the position shown in **Diagram 2**. The block then returns to the upright position as in **Diagram 1**.



Explain, in terms of centre of gravity, why the block returns to the upright position.

[2]

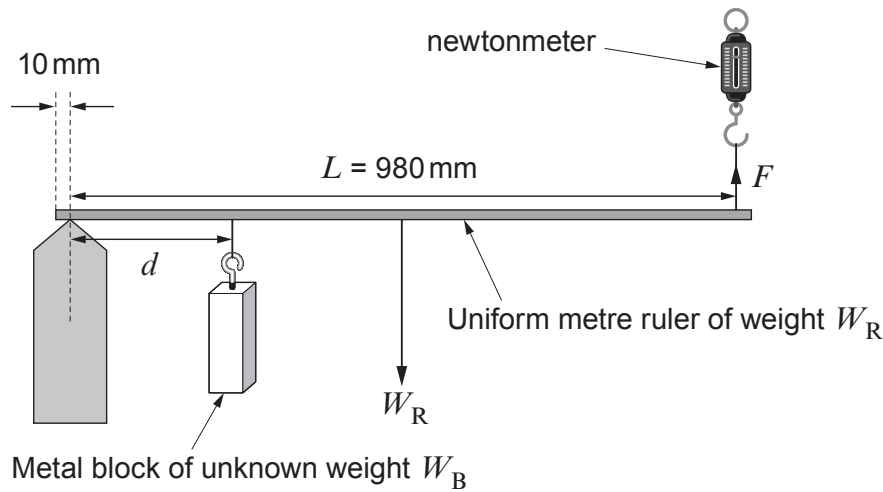
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(c) Sam sets up the experiment as shown using a metre ruler.



Sam uses the newtonmeter to measure the force, F , needed to make the metre ruler horizontal for varying values of d . He carries out the experiment twice and determines a mean value for F for each of the distances d . Sam judges that the ruler is horizontal by eye before taking readings for F . The measurements are recorded in the table below.

Distance, d / mm	Force, F / N		
	Trial 1	Trial 2	Mean
100	2.8	2.6	
250	5.2	5.7	
400	8.8	8.5	
550	11.2	11.6	
700	14.6	14.5	
850	17.2	17.6	

- (i) Complete the table. [1]
- (ii) The following relationship is used to find the weight, W_B , of the block:

$$F = \frac{W_B d}{L} + \frac{W_R}{2}$$

By taking moments about the pivot, show how this relationship is obtained. [2]

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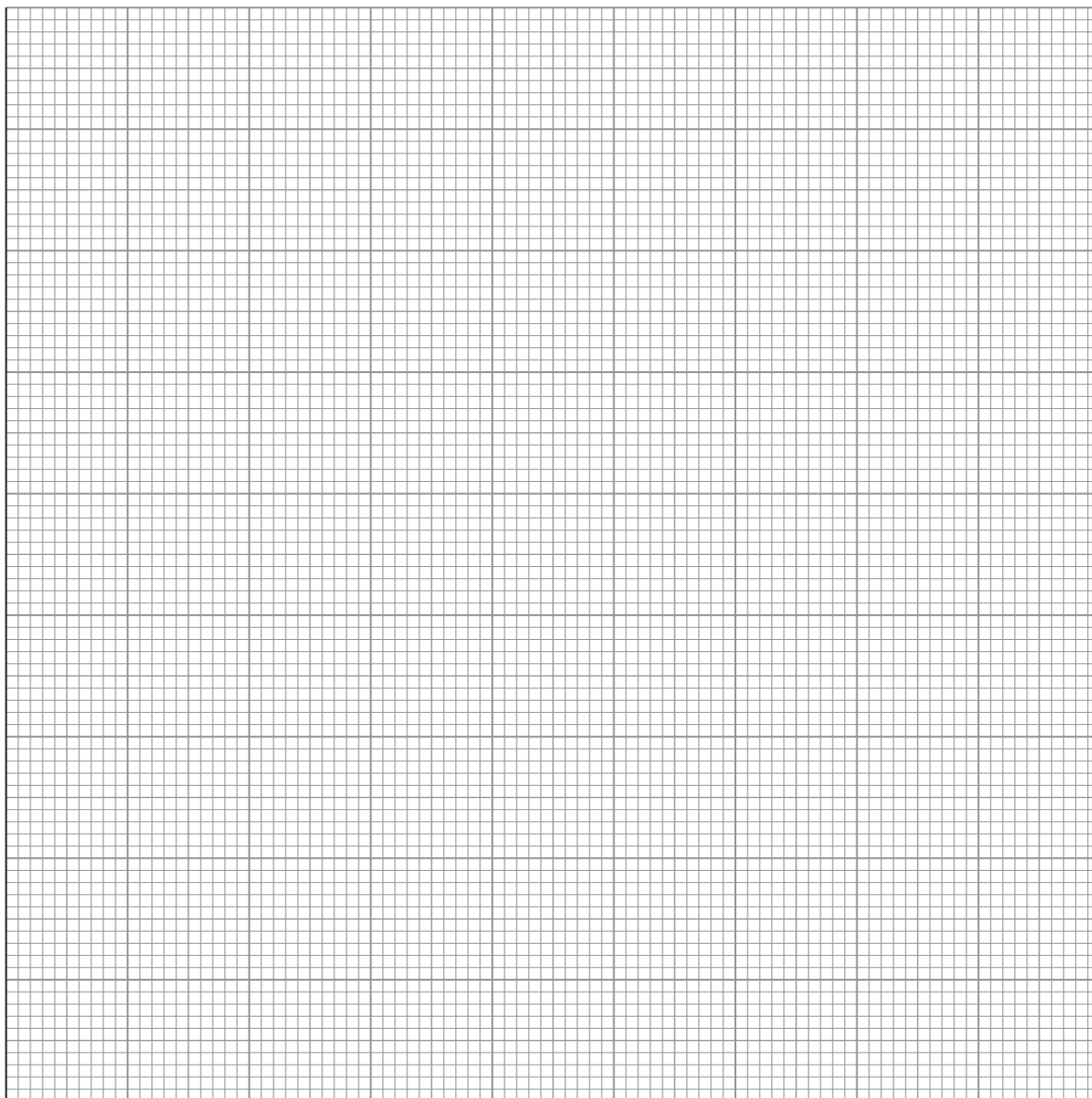
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- (iii) Use the grid to plot a graph of mean F (vertical axis) against d (in mm, on the horizontal axis) and draw a line of best fit. **Error bars should not be included.** [4]



(iv) Use your graph to determine:

I. W_B , the weight of the metal block;

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II. W_R , the weight of the ruler.

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(d) The distances, d , were measured to a resolution of ± 1 mm. Use data from the table of results to show that the percentage uncertainty in mean F is greater than the percentage uncertainty in d , when $d = 100$ mm. [3]

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(e) Suggest **one** change to his method which would improve the quality of Sam's measurements. [1]

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